

Chloe Barker

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EDUCATION

Southern Methodist University | Dallas, TX Jan 2025 - Present
M.S. in Data Science (Expected August 2026)

Dallas College | Dallas, TX Aug 2024 - Dec 2024
• Intro to Python Programming, Calculus I, Statistical Methods

The University of Texas at Austin | Austin, TX Aug 2020 - May 2024
B.S.A in Biology

Minor in Health Communication | Pre-Health Professions Certificate | GPA: 3.77
• Elements of Data Science, Biostatistics, Management Information Systems, Accounting, Health Communication, Physics I & II, Organic Chemistry I & II

TECHNICAL SKILLS

Programming & Analysis: Python (pandas, NumPy, scikit-learn, matplotlib), R (tidyverse, caret, ggplot2), SAS, SQL (MySQL, query optimization, joins, aggregations), Git/GitHub

Machine Learning & Modeling: Predictive Modeling, Classification (Logistic Regression, Random Forest, KNN, LDA/QDA), Regression (Linear, LASSO), Time Series, Model Selection (AIC/BIC), Feature Engineering, Cross-Validation, Model Evaluation (ROC/AUC, confusion matrices)

Data Management: Database Design & Normalization, ETL Workflows, Data Cleaning & Transformation, Relational Schema Design, Query Optimization

Statistics: Statistical Analysis, Hypothesis Testing, ANOVA, A/B Testing, Experimental Design

Visualization & Communication: Tableau, Power BI, R Shiny (Interactive Apps), ggplot2, matplotlib, seaborn, HTML/CSS, Executive Presentations, Technical Documentation, Data Storytelling

Tools: Jupyter Lab/Notebooks, VS Code, MS Office Suite (Excel, PowerPoint, Word, Visio), Google Suite

Data Platforms: Databricks, MySQL

Domains: Healthcare Analytics, Biological Data, Public Health Data (CDC, CMS, EPA, NIH)

WORK EXPERIENCE

Applied Analytics Group, LLC | Remote Jan 2026 - Present
Data Scientist

- Build predictive models and analytics pipelines using Python, R, and SQL to generate client-ready insights.
- Develop dashboards and automated workflows (Tableau, Power BI, Databricks) to support data-driven decisions.

Outlier AI | Remote Aug 2024 - Jan 2025
AI Math Evaluator

- Evaluated AI mathematical reasoning across algebra, calculus, and statistics to improve model accuracy.
- Designed structured test cases and error analyses to enhance reliability of reasoning-based AI systems.

RUSH University, Department of Microbial Pathogens and Immunity | Chicago, IL Jun 2023 - Aug 2023
Research Assistant

- Analyzed microbiome and immune-response datasets using Excel and R to study alcohol and circadian effects.
- Cleaned, visualized, and documented experimental data to support reproducible biomedical research.

The University of Texas at Austin, Nutrigenomics Biological Sciences Lab | Austin, TX Aug 2021 - Dec 2021
Undergraduate Researcher

- Investigated effects of artificial sweeteners on bacterial diversity using microbiome data and statistical methods.
- Conducted trend analysis and visualizations in Excel and R to communicate findings to research teams.

RUSH University, Department of Microbial Pathogens and Immunity | Chicago, IL Jul 2021 - Aug 2021
Research Assistant

- Analyzed longitudinal datasets on HIV medication impacts using descriptive statistics and data cleaning.
- Built visual summaries and maintained precise documentation supporting ongoing clinical research.

LEADERSHIP & COMMUNITY

Zeta Tau Alpha | *Philanthropy Committee, STAR Team, Member*

Phi Gamma Nu Business Fraternity: Epsilon Beta | *Member*

INTERESTS

Epidemiology, Healthcare Analytics, Running, Outdoor Sports, Psychology, Cooking, Strategy Games, Stand-Up Comedy